AI guided cellular rejuvenation research

Investor overview

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We are developing a research program to discover the causal mechanisms that allow adult cells to recover youthful function while retaining their identity. Our long-term objective is comprehensive reversal of cellular biological aging. The proposed first program would test a controllable mechanism in human skin cells and reconstructed skin, with tissue repair as a provisional therapeutic direction.

The opportunity

Artificial intelligence can integrate molecular measurements, nominate regulatory interventions and choose the next informative experiment. The commercially valuable outcome is a reproducible intervention with functional benefit, a usable delivery approach and clean rights. Our intended data advantage is a longitudinal record connecting actions to function, identity, safety and failure across donors.

The present evidence

An earlier research framework reports a treatment-expression signature in 34 public-data samples from three donors and six synthetic benchmark settings. The reported treatment-classification AUROC of 1.00 is not an efficacy measure. No biological rejuvenation result, validated proprietary target, clinical treatment or lifespan extension is claimed. The proposed financing would fund prospective biological discovery and independent confirmation.

The technical approach

We will benchmark multimodal cell representations, regulatory sequence models, graph and dynamic models, causal inference, imaging and Bayesian experiment selection. Language-model agents support source retrieval, reproducible analysis and execution tracking. Simple models and expert nomination remain competing baselines. New technologies are adopted when they improve confirmed discoveries per dollar.

We will begin with a planned 180-target screen in cells from eight adult donors, then confirm selected mechanisms in 16 new donors and two laboratories. Functional endpoints, cell identity, integrity and persistence after withdrawal determine advancement. Clocks are supporting measurements. Candidate mechanisms must withstand orthogonal perturbation and rescue tests.

Proposed financing

An illustrative $8.0 million, 24-month plan includes $7.4 million of operating work and a $0.6 million reserve. It funds discovery and lead feasibility for one tissue program. Clinical trials, GMP manufacturing and full IND-enabling development require later financing. The cash plan assumes zero commercial receipts.

Milestones commercial path and investor fit

Time	Planned evidence	Decision
First 3 months	Rights review, experimental leader, computational replay, qualified functional assay and quotes	Approve the full screen or narrow the plan
Months 4 to 12	Prospective AI versus expert versus random selection and functional confirmation	Select up to two mechanisms for deeper study
Months 13 to 18	New-donor and external-laboratory replication with withdrawal observations	Select one lead mechanism if the evidence supports it
Months 19 to 24	Delivery feasibility, IP assessment and partner diligence package	Partner, finance the next study or discontinue

The first commercial route is a defined research collaboration or option agreement with a biotech or pharmaceutical partner. Up to two illustrative $0.5 million pilots are a business-development objective, not signed revenue. Asset licensing requires confirmed biology and a defensible rights position. We do not base the seed financing on royalties, an acquisition or consumer anti-aging sales.

Why this could become defensible

Widely available AI models reduce the value of model access alone. A validated intervention, prospective human-cell response data, independent replication and knowledge of failure contexts are harder to reproduce. These assets must be created by the funded program. A broad claim to AI-driven rejuvenation is not sufficient intellectual property.

Principal risks

The program may produce clock changes without meaningful function, benefits that disappear after withdrawal, loss of cell identity, abnormal growth, weak donor generalization or no advantage over expert selection. Costs can exceed the plan and buyers may decline paid collaborations. Each outcome has a predefined decision to narrow or stop. Short-term safety observations cannot exclude delayed cancer risk.

Team and formation

The named authors are David Xu, PhD; Peter Lee, PhD, MD; and Zhimin Zheng, PhD. Formal company roles, time commitments, ownership and institutional permissions are financing diligence items. Experienced experimental and translational leadership must be appointed. No institutional sponsorship or proprietary biological rights are asserted through authorship.

Discussion requested

We seek investors with a demonstrated interest in longevity biology, regenerative medicine, early therapeutic discovery or AI-enabled experimental science. The initial discussion is about scientific fit, team formation and a milestone-financed seed program. Valuation, structure and securities terms would be addressed separately.

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The accompanying technical white paper provides the complete experiment plan, budget, risks, investor questions and 21 references. Planning figures are assumptions; no therapeutic success is promised.